Adapting the MEE Pipeline to PUNCH Coronagraph Data

Centroid validation, Gaia matching, and a refined measurement strategy

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Executive Summary

I adapted the MEE2024 Tab 1 centroid pipeline to PUNCH Level-3 white-light images and built a Gaia-based validation workflow using PUNCH's mission WCS (the built-in plate solve does not converge on coronagraph data).

Centroid detection and Gaia matching work well (86–88% purity). However, the stars I can reliably detect sit far from the Sun (nearest confirmed match ~84 solar radii), where the gravitational deflection signal is ~1000x below our position noise. A measurement of the Einstein coefficient L is therefore not achievable with this data as-is.

I am now pivoting to a visual-first strategy: confirm stars by eye (DS9) before trusting automated matches, which improves reliability and sets up a cleaner multi-image test of the method.

1. Data & Setup

Instrument: PUNCH (Polarimeter to Unify the Corona and Heliosphere), Level-3 white light.
Frame: 2026-02-19 00:16 UTC, 4096x4096, Sun centered at pixel (2048, 2048).
Plate scale: ~81 arcsec/pixel — about 44x coarser than MEE eclipse data (~1.85 arcsec/px).
Implication: Each pixel covers a large patch of sky; faint/close stars blur and the corona dominates, producing many false centroids that must be filtered.

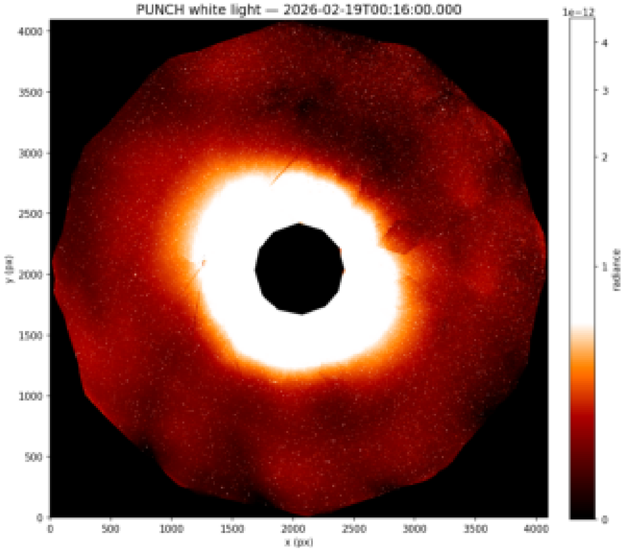

Figure 1. PUNCH L3 white-light frame (2026-02-19). Stars appear as faint points across a corona-dominated field.

2. What I Did — and the Limiting Result

2.1 Centroid finding + Gaia validation

I ran Tab 1 to detect centroids, then projected the Gaia catalog onto the image through PUNCH's WCS and matched detections to catalog stars. Two representative runs:

Run	Centroids	Matched	Purity	Completeness
Full field (corona-masked)	4,343	3,749	86.3%	44.8%
Top-50 brightest (G<4)	50	44	88.0%	45.4%

Takeaway: the detector reliably finds real stars (high purity); roughly half the catalog stars in-field are recovered.

2.2 The limiting result: deflection vs. noise

Closest confirmed star:	~84 solar radii from Sun center.
Predicted deflection there:	~0.021 arcsec (~0.0003 pixels).
Measured position noise:	~23 arcsec (~0.28 pixels).
Signal-to-noise:	~0.001 — the signal is ~1000x below the noise floor.

This is a physics + resolution limit, not a tuning problem: the detectable stars are simply too far from the Sun, and the plate scale is too coarse, for the bending to register.

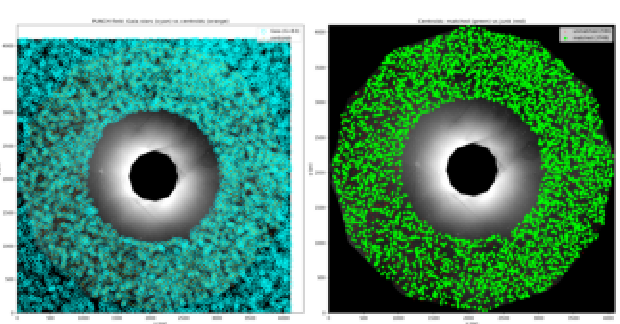

Figure 2. Gaia stars projected onto centroids (full-field run).

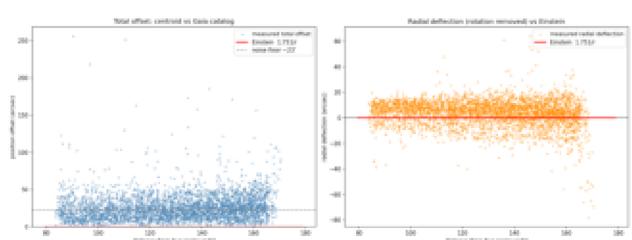

Figure 3. Measured offsets vs. tiny predicted deflection — noise dominates.

3. Refined Approach & Next Steps

3.1 Visual-first validation

Rather than trusting the brightest detections outright (some are corona features, not stars), I moved to a visual-first workflow: examine the frame in DS9, confirm which point sources are genuinely star-like, and only then match those to Gaia. A helpful conversation with Henry Throop (NASA) reinforced this direction — he suggested confirming stars by eye before trusting automated matches, which I have folded into the process.

I built a Top-20 comparison (20 brightest observed centroids vs. 20 brightest predicted Gaia stars) for direct visual inspection. It confirms the geometry issue: several very bright centroids near the Sun do NOT correspond to Gaia stars (they are corona structure), while the nearest visually-confirmed matches lie at ~98–160 solar radii.

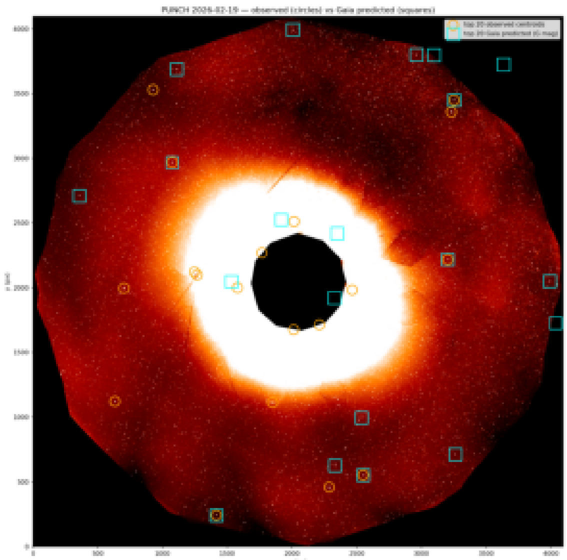

Figure 4. Top-20 observed centroids vs. Top-20 predicted Gaia stars (visual inspection, no blind auto-match).

3.2 A multi-image idea (and an honest caveat)

Extending this across several PUNCH dates would let me track the same Gaia-confirmed stars and test whether offsets follow the expected 1/R law. Averaging N epochs reduces random error as ~1/sqrt(N). Caveat: with the current ~1000x signal-to-noise gap, this characterizes our error budget and validates the method — it will not by itself yield a real L measurement until stars can be detected much closer to the Sun (e.g., finer plate scale such as STEREO COR1, or eclipse data).

Next steps

- DS9 visual inspection of PUNCH frames to establish a trusted star list.
- Targeted Gaia matching on visually-confirmed stars (not blind brightest-N).
- Extend Top-20 / Gaia overlay to multiple dates; measure offset scatter vs. solar radius.
- Evaluate finer-scale references (STEREO COR1) to test the workflow where stars sit closer in.